Inverse-fiber primitive run record

2026-10-16 · Native Rust · Darwin L0 and GitHub-hosted Linux L1

The prespecified experiment is `PROTOCOL_V1.md`. The public input is `order1000/BATCH_1237_1201_V2.public.json`, SHA-256 `24f2c56a9ba56f91c954f38b7752edf24fee5ed027b49b1ac0747771945099c4`. The historical solver is `inverse_velu/FROZEN_RUNNER_V2.rs`, SHA-256 `7587fc43ffbfe6042c3915f90d4c7dc164337fce61d0599776f2b607150660c3`. The instrumented source byte copy is `FROZEN_RUNNER_V1.rs.txt`, SHA-256 `0731ddc40f26397fdd30bed5098613a3bff91ec3c74c3863c7079d0313b8a250`.

Run command from the repository root:

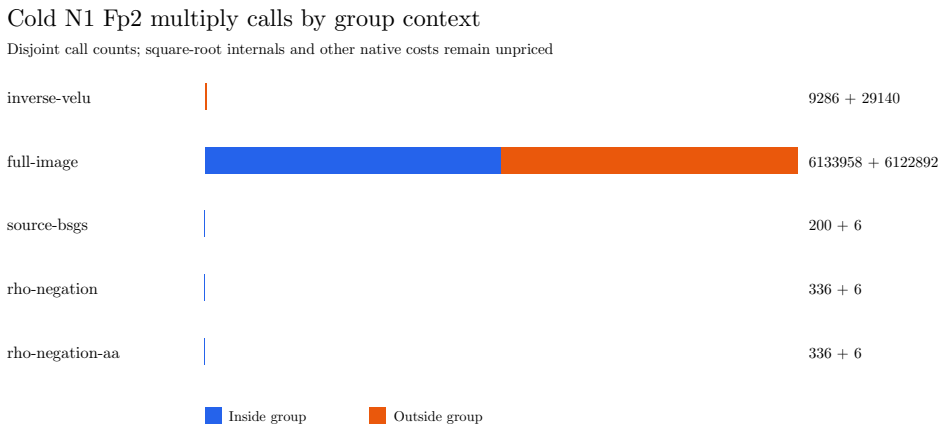
```
bash research/isogeny_torsion_fibers_20261012/primitive_price_20261016/run_panel_v1.sh research/isogeny_torsion_fibers_20261012/primitive_price_20261016/RUN_V1
```

The shell launcher ran five arms at prefixes 1, 4, 16 and 64 over three rotated rounds. `RUN_V1/raw/` retains each child stdout, stderr, exit, meter stdout and meter JSON. `RUN_V1/build-sha256.txt` binds the binary, sources, inputs and script. `RUN_V1/ARTIFACT_FILES.sha256` verified after the run. All 60 exits were zero; `RUN_V1/replay.json` reports 1,275 scalar and 6,126 relation-witness checks passed. The native reduction command was:

```
isogeny_fiber_primitive_price audit RUN_V1 inverse_velu/RUN_V2 PRIMITIVE_GRAPH_V2.svg
```

Its V2 output compared all 60 deterministic old/new records. The first audit attempt failed because its reducer assumed rho had a `query_group_ops` field. The actual field is `rho_group_ops`; the failed attempt and corrected V2 output are both preserved beside this report. The reducer error did not change the measured children.

Independent Linux artifacts came from PR 1939 runs 38961930762 (60-child inverse panel), 38961930725 (108-child nine-arm panel), and 38961930860 (hard-capped order-2,437 full-image probe). All copied artifact manifests verified. The first two replay checkers passed. The full-image child exited 134 after an allocation failure under the 512 MiB address-space cap, leaving a resource-censored row.



This graph records disjoint top-level Fp^2 multiplication calls at N1. Square-root internals, polynomial/table control flow, allocations, I/O and independent-host calibration remain unpriced. The result PDF and `RESULTS_V1.md` give the per-arm counts and exact limits.